

# Fundamental theorem of Calculus Part 1

this theorem is taking the derivative of a integral

ex #1:

$$\int_0^x (t^2 + 1)$$

answer:

$$f'(x) = x^2 + 1$$

For most of the problems you will have something like this all you have to do is replace the "t" with the "x"

ex #2:

$$\int_x^0 (t^3 + 1)$$

answer =

$$-\int_0^x (t^3 + 1)$$

$$f'(x) = -(x^3 + 1)$$

the "x" or whatever letter it is must be on top you will need to use some integral properties

ex #3:

$$\int_0^{x^2} (t^4 + 1)$$

answer:

$$f'(x) = ((x^2)^4 + 1) \cdot \frac{d}{dx} [x^2]$$

$$f'(x) = (x^8 + 1) \cdot 2x$$

Now there isn't a "x" on top there is an "x<sup>2</sup>" which means you need to do the same exact thing you have been doing just take the derivative of the term or variable that is on top of the integral and multiply it by the rest

anytime there is something other than an "x" you will do the same thing but take the derivative of it and multiply it by the rest of the answer

$$\int_{x^2}^{x^2-1} (t^2+1)$$

When you encounter something like this you must split it into two parts

$$\int_{x^2}^0 (t^2+1) + \int_0^{x^2-1} (t^2+1)$$

(x<sup>2</sup> is not on top you need to get it on top)

$$-\int_0^{x^2} (t^2+1) + \int_0^{x^2-1} (t^2+1)$$

$$F'(x) = -((x^2)^2+1) \cdot \frac{d}{dx}[x^2] + ((x^2-1)^2+1) \cdot \frac{d}{dx}[x^2-1]$$

$$F'(x) = -2x(x^4+1) + 2x((x^2-1)^2+1)$$

# Fundamental theorem of calculus Part 2

The Fundamental theorem of calculus Part 2 will look like this for  $b$  and  $a$  these are numbers

$$\int_a^b f(x) dx \longrightarrow f(b) - f(a)$$

you must integrate the  $f(x)$  First they place  $b$  and  $a$  in they must be subtracted

ex:

$$\int_2^3 \left( \frac{5}{x^2+1} + \frac{2}{x^2} \right)$$

break it up using the integral Properties to make it easy to deal with the 5 and two can be pulled out because they are a constant

$$5 \int_2^3 \frac{1}{x^2+1} + 2 \int_2^3 \frac{1}{x^2}$$

$$5 \left[ \tan^{-1}(x) \right]_2^3 + 2 \left[ -\frac{1}{x} \right]_2^3$$

the number on top are used to remind you that you still need to plug them in

$$5 \left[ \tan^{-1}(3) - \tan^{-1}(2) \right] + 2 \left[ -\frac{1}{3} + \frac{1}{2} \right]$$

$$5 \left[ \tan^{-1}(3) - \tan^{-1}(2) \right] + \frac{1}{3}$$

antiderivative of  $\int \sec x \cdot \tan x = \sec x$

$$\sec = \frac{1}{\cos x}$$

$$\int_0^{\pi/4} \sec x \cdot \tan x \, dx$$

$$\left[ \sec x \right]_0^{\pi/4} \Rightarrow \sec\left(\frac{\pi}{4}\right) - \sec(0)$$

$$\frac{1}{\cos(\pi/4)} - \frac{1}{\cos(0)} = \boxed{-1 + \sqrt{2}}$$

$$\int_0^{\pi/4} (10 \sec x \tan x + 3 \sec^2 x) \, dx$$

$$10 \int_0^{\pi/4} \sec x \tan x + 3 \int_0^{\pi/4} \sec^2 x$$

$$10 \left[ \sec x \right]_0^{\pi/4} + 3 \left[ \tan x \right]_0^{\pi/4}$$

$$10 \left[ \frac{1}{\cos(\pi/4)} - \frac{1}{\cos(0)} \right] + 3 \left[ \frac{\sin(\pi/4)}{\cos(\pi/4)} + \frac{\sin(0)}{\cos(0)} \right]$$

$$10(-1 + \sqrt{2}) + 3(1)$$

$$-10 + 10\sqrt{2} + 3 = \boxed{-7 + 10\sqrt{2}}$$